

05) Polynomial Regression

5

$$R^2 = 0.94$$

$$p = 3$$

$$n = 10$$

$$R^2_{adj} = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - p - 1} \right]$$

$$= 1 - \left[\frac{(1 - 0.94) \cdot 9}{10 - 3 - 1} \right]$$

$$= 1 - \frac{0.54}{6} = 1 - 0.09$$

$$\underline{R^2_{adj} = 0.91}$$

6) ~~05)~~ The result is — B

7) Adjusted R^2 is when we want to check the ~~efficiency~~ efficiency of number of features as it gives penalty for each feature ~~added~~ added, in contrast for R^2 which each feature will improve overall score, $adj R^2$ gives us the scope to see if the model is overfitted or not, and gives us true cost of added features